Quantification: 535_z20c1-2
==========================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	27
interior lacuna count	20
frame-edge lacuna count	7

Interior lacunae only: mean, median and sample SD (n = 20)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2999.35	2390.5	2408.95	px^2
major_axis_length_px	lacuna shape	87.307	84.2904	31.394	px
minor_axis_length_px	lacuna shape	45.9626	40.768	22.2471	px
aspect_ratio	lacuna shape	2.1118	1.9872	0.7654	unitless
eccentricity	lacuna shape	0.8248	0.8639	0.1271	unitless
solidity	lacuna shape	0.787	0.7953	0.1064	unitless
roots_count	headline	8.35	7.5	4.1074	count
ring_length_r30_px	headline	263.85	228	116.04	px
ring_length_r60_px	reported	559.05	490.5	255.891	px
owned_length_px	ownership-dependent	628.301	556.487	386.874	px
edge_count	ownership-dependent	28.75	23	18.2609	count
mean_edge_length_px	ownership-dependent	22.5335	21.3456	4.3968	px
perimeter_px	normalised	295.408	252.664	159.555	px
ring_area_r30_px2	normalised	9145.7	8613.5	3600.85	px^2
ring_area_r60_px2	normalised	19709.5	17576	6799.67	px^2
in_frame_fraction_r30	normalised	0.9875	1	0.0408	unitless
in_frame_fraction_r60	normalised	0.9753	1	0.0755	unitless
ring_density_r30	normalised	0.0287933	0.0266877	0.00566848	px^-1
ring_density_r60	normalised	0.0279167	0.0294762	0.00628226	px^-1
roots_per_100px_perimeter	normalised	3.0181	3.0634	1.0352	per 100 px
ring_attached_length_r30_px	network v2	226.85	201.5	117.305	px
ring_attached_length_r60_px	network v2	409	334.5	245.19	px
ring_length_w_r30_px	network v2	298.898	256.122	131.704	px
ring_length_w_r60_px	network v2	635.02	561.215	294.494	px
sholl_crossings_r10	network v2	6.45	6	3.3478	count
sholl_crossings_r20	network v2	7.65	7.5	4.1584	count
sholl_crossings_r30	network v2	7.45	6.5	3.6631	count
loop_count	loops, report only	0.9	0	1.7442	count
loop_edge_share	loops, report only	0.0656	0	0.1235	share
ring_width_mean_r30_px	width	7.8577	7.5841	1.381	px

Per field
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measure	value	units
analysed_area_px2	975704	px^2
total_skeleton_length_px	21850	px
canalicular_length_density_per_px	0.0223941	px^-1
junction_count	367	count
junction_density_per_px2	0.00037614	px^-2
lacuna_count	27	count

lacunae_per_px2	2.767e-05	px^-2
skeleton_component_count	380	count
mean_component_length_px	57.5	px
median_component_length_px	22	px
field_length_density_w_per_px	0.0252278	px^-1
field_density_without_flagged_per_px	0.0220547	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	586	px
unattached_component_count	36	count
canalicular_length_density_attached_per_px	0.0223941	px^-1
field_length_density_with_unattached_per_px	0.0229947	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	25	count
loop_edge_share	0.0887	share
edge_count	1105	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.5574	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	11.6619	px
width_pixels_used	19238	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	766	55.3886	19.9792	2.7723	0.9327	0.8815	1.5239	7.094	208.97	True
2	1417	59.2424	36.6108	1.6182	0.7862	0.7182	0.2448	34.1404	249.047	False
3	492	40.0308	17.439	2.2955	0.9001	0.8571	-1.4757	7.689	940.191	True
4	2789	96.2462	44.7487	2.1508	0.8853	0.7481	1.2422	44.701	163.562	False
5	3380	83.9727	56.9872	1.4735	0.7345	0.8272	-0.9852	92.8941	716.001	False
6	532	31.4193	21.8844	1.4357	0.7175	0.9449	-0.5795	222.887	852.859	False
7	1982	85.686	32.1789	2.6628	0.9268	0.8251	-1.1473	243.598	120.229	False
8	670	34.0595	26.8161	1.2701	0.6165	0.9293	0.3429	300.875	1012.39	True
9	1956	84.6082	30.6094	2.7641	0.9323	0.8815	-0.4607	416.865	210.624	False
10	7294	147.859	76.1316	1.9421	0.8573	0.7458	-0.0651	474.999	284.411	False
11	4856	115.922	57.0373	2.0324	0.8706	0.8161	-1.3108	461.194	574.536	False
12	3706	78.2414	69.9705	1.1182	0.4475	0.789	0.5114	581.897	502.58	False
13	2088	68.5717	40.0641	1.7115	0.8116	0.9198	0.595	570.565	144.556	False
14	2662	104.173	41.4719	2.5119	0.9173	0.6642	1.1534	613.184	873.72	False
15	10346	169.284	102.254	1.6555	0.797	0.627	-0.6171	678.698	677.243	False
16	2280	82.4791	35.9781	2.2925	0.8998	0.9205	-1.1889	632.433	317.15	False
17	1049	49.7107	28.8293	1.7243	0.8147	0.9468	1.1742	770.511	315.037	False
18	524	55.7224	13.3728	4.1669	0.9708	0.8397	-1.4869	765.378	856.048	False
19	4401	92.4669	72.9809	1.267	0.6141	0.7373	1.4761	819.541	677.56	False
20	4198	98.8193	64.1556	1.5403	0.7606	0.7155	1.4566	847.201	603.76	False
21	2501	85.9153	42.1925	2.0363	0.8711	0.8016	1.0022	846.978	318.042	False
22	1180	74.8721	28.9962	2.5821	0.922	0.7121	0.5457	855.245	269.011	False

23	1473	60.4844	33.1719	1.8234	0.8362	0.8731	-0.2053	858.547	1008.1	True
24	846	80.9285	22.7986	3.5497	0.9595	0.5599	-0.1838	864.301	863.892	False
25	703	38.461	24.103	1.5957	0.7793	0.9083	-0.1158	967.855	1012.53	True
26	3758	122.147	46.3178	2.6372	0.9253	0.7952	-1.5021	1003.64	793.015	True
27	5023	142.163	51.4092	2.7653	0.9323	0.803	-1.5486	1001.47	676.141	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	0	12	12	0	0	0	
2	5	126	195	308.735	18	17.1519	
3	4	81	264	183.882	7	26.2689	8.1954
4	10	233	493	587.002	31	18.9356	12.124
5	7	351	834	680.671	23	29.5944	7.4729
6	5	203	544	525.973	23	22.8684	7.5355
7	9	398	928	1231.25	59	20.8686	8.2992
8	1	61	362	200.024	10	20.0024	10.932
9	7	259	488	676.5	31	21.8226	9.2021
10	16	427	813	1139.9	62	18.3855	8.3815
11	11	290	649	791.127	31	25.5202	8.3643
12	11	391	733	791.281	40	19.782	7.96
13	7	227	616	354.865	19	18.6771	6.293
14	10	298	655	689.027	26	26.501	9.4006
15	20	588	1250	1748.72	72	24.2877	7.2058
16	7	229	366	330.735	17	19.455	8.07
17	2	156	323	414.534	15	27.6356	7.5841
18	6	151	387	434.635	14	31.0453	7.8686
19	8	202	305	326.451	16	20.4032	6.5905
20	8	200	336	376.794	15	25.1196	5.8269
21	8	203	442	749.742	46	16.2987	6.8227
22	3	144	455	128.569	7	18.3669	6.878
23	5	140	262	511.233	31	16.4914	7.4904
24	7	201	369	279.51	10	27.951	7.4164
25	4	99	229	43.9706	4	10.9926	7.8076
26	6	169	324	233.066	15	15.5377	8.1017
27	5	133	299	91.598	7	13.0854	4.4072

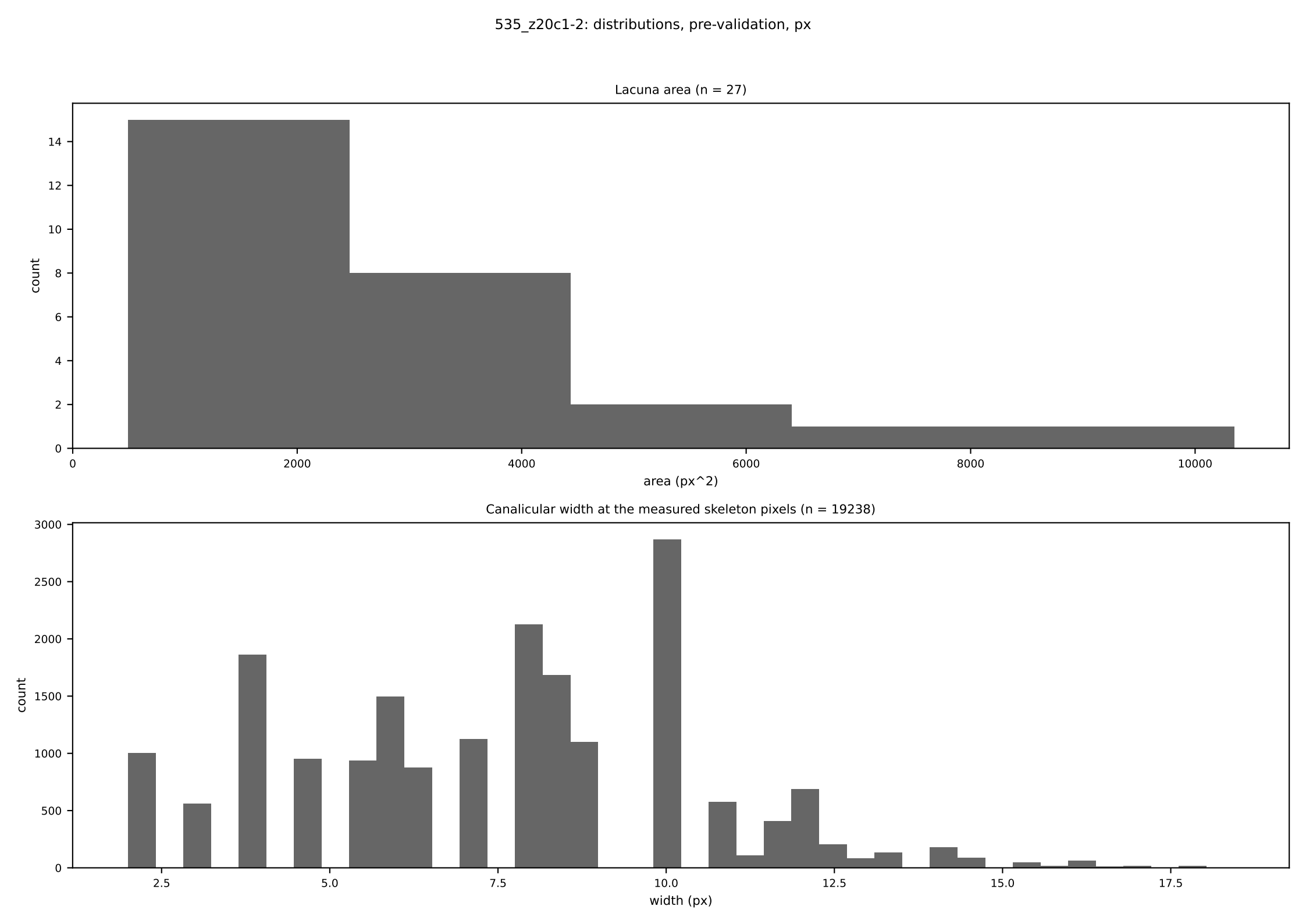
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	153.326	888	888	0.5445	0.5303	0.0135135	0.0135135	0
2	226.072	4957	10553	0.8341	0.734	0.0254186	0.0184782	2.2117
3	101.941	3528	9806	0.6148	0.5733	0.0229592	0.0269223	3.9238
4	325.848	7077	13698	0.915	0.7774	0.0329236	0.0359907	3.0689
5	303.605	10283	25801	1	0.9954	0.034134	0.0323243	2.3056
6	92.8112	5363	16377	1	1	0.037852	0.0332173	5.3873
7	243.179	9488	24489	1	1	0.0419477	0.0378946	3.701
8	106.184	3521	9851	0.5977	0.5656	0.0173246	0.0367475	0.9418
9	208.894	7431	16405	1	1	0.034854	0.029747	3.351
10	446.174	13512	24714	1	1	0.0316015	0.0328963	3.586
11	391.931	12275	26441	1	1	0.0236253	0.0245452	2.8066
12	312.191	11238	23739	1	1	0.0347927	0.0308775	3.5235
13	198.995	8411	21092	1	1	0.0269885	0.0292054	3.5177
14	327.019	11681	28179	1	1	0.0255115	0.0232443	3.0579
15	801.063	20617	37918	1	1	0.0285202	0.0329659	2.4967
16	223.965	8975	20144	1	1	0.0255153	0.0181692	3.1255
17	139.681	5912	12774	1	1	0.026387	0.0252857	1.4318

18	126.083	6387	15433	1	1	0.0236418	0.0250761	4.7588
19	473.073	8816	14538	1	1	0.0229129	0.0209795	1.6911
20	421.339	9498	18437	1	1	0.0210571	0.0182242	1.8987
21	262.149	6917	12874	1	1	0.029348	0.0343328	3.0517
22	206.309	6064	13870	1	1	0.0237467	0.0328046	1.4541
23	173.539	4894	10662	0.636	0.5937	0.0286065	0.0245733	2.8812
24	177.782	8012	16715	1	1	0.0250874	0.022076	3.9374
25	112.912	3718	7367	0.6229	0.5173	0.0266272	0.0310846	3.5426
26	380.143	5868	11373	0.6218	0.5939	0.0288003	0.0284885	1.5784
27	445.605	6941	13590	0.6459	0.6098	0.0191615	0.0220015	1.1221

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	6	6	13.4853	13.4853	1	0	0
2	108	111	146.297	225.894	3	2	2
3	75	105	90.0122	293.037	3	1	5
4	103	222	260.622	558.801	5	8	6
5	330	651	400.676	954.992	8	14	8
6	184	476	239.279	627.985	4	9	8
7	375	815	452.433	1047.05	11	10	11
8	61	135	66.6274	413.505	1	2	1
9	259	449	298.593	563.63	6	7	9
10	353	563	478.99	918.394	8	16	14
11	250	478	320.593	728.784	12	9	7
12	332	663	445.99	846.666	8	13	12
13	125	217	255.693	689.257	6	7	10
14	220	428	343.948	741.683	6	11	10
15	559	1068	666.227	1439.58	16	12	15
16	226	356	256.551	415.505	6	8	6
17	89	182	176.782	360.764	3	7	5
18	126	217	166.811	431.049	3	2	5
19	198	265	227.924	344.35	6	3	3
20	200	259	219.782	367.108	5	3	3
21	203	313	238.723	507.617	3	5	5
22	100	146	162.51	524.458	5	3	6
23	139	243	164.196	297.208	3	3	6
24	197	301	219.539	406.836	5	4	4
25	74	78	108.012	252.409	2	1	2
26	137	208	189.681	366.605	5	6	3
27	78	82	141.983	326.693	4	4	4


535_z20c1-2: distributions, pre-validation, px
Lacuna area (n = 27)
count
area (px^2)
0
2
4
6
8
10
12
14
0
2000
4000
6000
8000
10000
Canalicular width at the measured skeleton pixels (n = 19238)
count
width (px)
0
500
1000
1500
2000
2500
3000
2.5
5.0
7.5
10.0
12.5
15.0
17.5